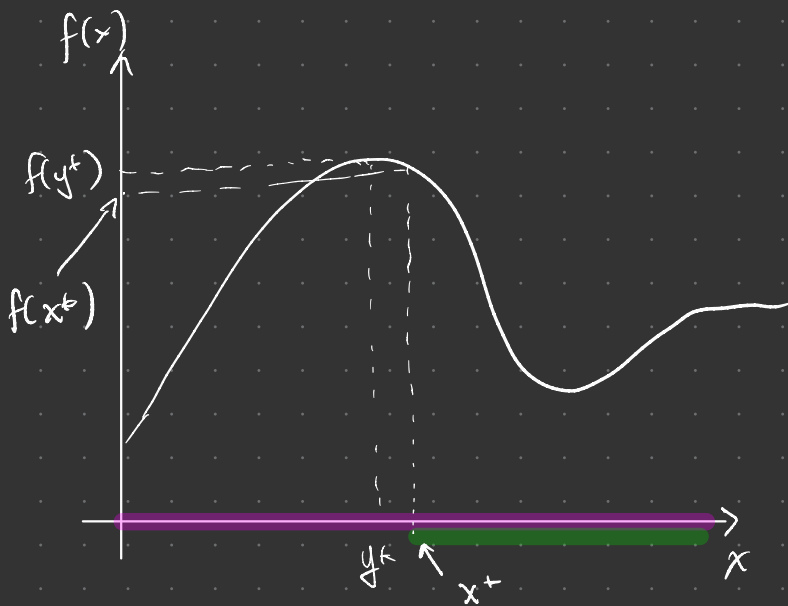


X
 Y

$$X \subseteq Y$$

$$\max_{x \in Y} f(x)$$

$$\max_{x \in X} f(x)$$

$$f(y^*) \geq f(x^*)$$

$$\lambda [P_T - c] + (1-\lambda) [P_B - c] = \lambda P_T + (1-\lambda) P_B - (\lambda c + (1-\lambda)c)$$

$$= \lambda P_T + (1-\lambda) P_B - c$$

↓
objective

$$E_{x.} \text{ Profit} = \lambda P_T + (1-\lambda) P_B - c$$

What if c is different depending on type $\{B, T\}$

$$E_{x.} \text{ Profit: } \lambda [P_T - c_T] + (1-\lambda) [P_B - c_B]$$

From 17 & 18

$$(17) \quad v = \theta_T P_T + w_T \quad \text{IRT or PCT}$$

$$(18) \quad \theta_B P_B + w_B = \theta_B P_T + w_T \quad (1C3)$$

$$\text{From (17)} \quad \frac{v - w_T}{\theta_T} = P_T \quad (19)$$

From (19) & (18) we get

$$\theta_B P_B + w_B = \frac{\theta_B}{\theta_T} (v - w_T) + w_T$$

$$\Rightarrow \theta_B P_B = \frac{\theta_B}{\theta_T} (v - w_T) + w_T - w_B$$

$$\begin{aligned} \Rightarrow P_B &= \frac{\cancel{\theta_B}}{\theta_T \cancel{\theta_B}} (v - w_T) + \frac{(w_T - w_B)}{\theta_B} = \frac{v - w_T}{\theta_T} + \frac{w_T - w_B}{\theta_B} \\ &= \frac{v - w_T}{\theta_T} + \frac{w_T}{\theta_B} \quad (20) \end{aligned}$$

From (c) we showed

that $w_B = 0$ at
an optimal solution

R.P.

$\max_{(w_T, w_B)}$

$$\lambda \left(\frac{v - w_T}{\theta_T} \right) + (1-\lambda) \left(\frac{v - w_T}{\theta_T} + \frac{w_T - w_B}{\theta_B} \right) - c$$

$$\text{s.t.} \quad 0 \leq w_T \leq v \\ w_B \geq 0$$

$\Rightarrow$ Since w_B appears as the term $-(1-\lambda) \frac{w_B}{\theta_B}$

$\Rightarrow w_B = 0$ at an optimal solution.

$$\max_{w_T \in [0, v]} \left\{ \left(\frac{\sigma}{\theta_1} - c \right) + w_T \left(\frac{1-\lambda}{\theta_3} - \frac{1}{\theta_1} \right) \right\}$$

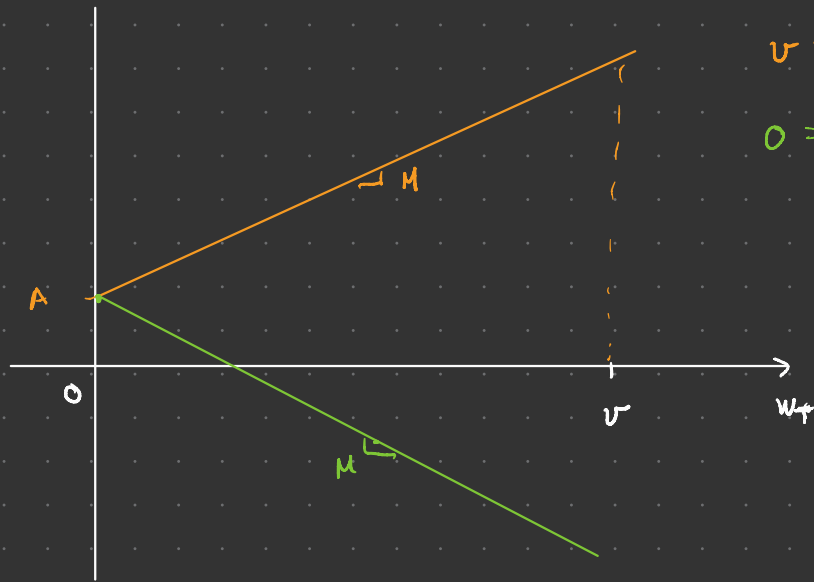
$$A + w_T M$$

where $A = \frac{\sigma}{\theta_1} - c$, $M = \frac{1-\lambda}{\theta_3} - \frac{1}{\theta_1}$

$$(23) \max_{w_T \in [0, v]} \{A + w_T M\}$$

A is the intercept

M is the slope



$$v = \operatorname{argmax} (23) \Rightarrow w_t^* = v \text{ if } M > 0$$

$$0 = \operatorname{argmax} (23) \Rightarrow w_t^* = 0 \text{ if } M < 0$$

$$-p(e_L - 1) \leq 0 \quad \text{with equality if } e_L > 0$$



$$-p(e_L - 1) + \lambda_L = 0, \quad \lambda_L \geq 0, \quad \lambda_L e_L = 0, \quad e_L \geq 0$$

$$\downarrow$$
$$-p(e_L - 1) = -\lambda_L \leq 0$$

Type From PS4- solutions: eq (27) should be

$$q_i = \max \left\{ \theta_i - \frac{1 - \pi_i}{\pi_i} (\theta_{i+1} - \theta_i), 0 \right\}$$

$$\theta_1 = 1, \quad \theta_2 = 5, \quad \theta_3 = 9$$

$$\pi_1 = \pi_2 = \pi_3 = 1/3$$

$$\pi_2 = \sum_{i=1}^2 \pi_i = 1/3 + 1/3 = 2/3$$

$$q_2 = \max \left\{ 5 - \frac{1 - (1/3 + 1/3)}{1/3} (\overbrace{9 - 5}^{\theta_3 - \theta_2}), 0 \right\}$$

$$= \max \left\{ 5 - \frac{1 - 2/3}{1/3} (9 - 5), 0 \right\}$$

$$= \max \left\{ 5 - \frac{1/3}{1/3} (4), 0 \right\} = 5 - 4 = 1$$